

Ollama & vLLM through the Open OnDemand Web Interface

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Partnership for an Advanced Computing Environment

PACE provides leading edge advanced research computing resources with technical support services, infrastructure software, and more.

Compute & Storage



PACE Team



Software Support



PACE Services

- ✓ CPU/ GPU Compute Capacity
- ✓ Individual & Shared Project Storage
 - Backup services
 - Globus
- ✓ Classroom & Teaching Resources
- ✓ Software
 - Open-source packages (Python, R, etc.)
 - Some licensed software
 - Singularity/Apptainer container support

- ✓ User Support
 - Workshops, Trainings (Linux 101/102, Python 101, Machine Learning, Using Containers, Optimization 101, etc.)
 - Schedule: https://gatech.servicenow.com/home?id=kb_article_view&sysparm_article=KB0042298
 - No-cost weekly consultations/ help sessions
 - Email support
 - pace-support@oit.gatech.edu
 - User documentation
 - https://gatech.servicenow.com/technology?id=kb_article_view&sysparm_article=KB0042503
 - Video recordings

Ollama v. vLLM

Similarities

- Easily run LLM, VLMs, etc from the terminal or Open OnDemand
- Apptainer images provided by PACE
- OpenAI-compatible chat and completion APIs are available for both Ollama and vLLM

Ollama v. vLLM

Differences

- vLLM for batch scripts and Ollama for prototyping
- vLLM...
 - Runs only one model at a time
 - Slow startup time (few mins.)
 - Lower latency and higher-throughput after startup
- Ollama ...
 - Runs multiple models at a time
 - Also available on the terminal: `module load ollama`

Chat and Completions APIs Demo

- Show the chat and completions APIs of Ollama and vLLM, all from the browser
- Both use the same OpenAI-compatible APIs
 - First prototype with Ollama, and run batches using vLLM



vLLM Custom Command-line Config. Demo

- Custom configuration for vLLM
 - Command-line arguments
 - Environment variables
 - E.g., `VLLM_FLOAT32_MATMUL_PRECISION={highest,high,medium}`
 - E.g., `VLLM_LOGGING_LEVEL`
- Specify a specific location for model storage
- Custom Apptainer arguments (e.g., `--nvccli`)

Ollama & vLLM through the Open OnDemand Web Interface

- Ollama and vLLM are available on JupyterLab via a few clicks on Open OnDemand
 - Try it out! ondemand-phoenix.pace.gatech.edu
- vLLM for batch scripts and Ollama for prototyping
 - vLLM provides lower-latency and higher-throughput responses
 - Ollama supports switching models quickly and starts quickly
- Both chat and completion APIs are available for both Ollama and vLLM
- Ollama has been updated to ver. 0.12.11
 - Can now run latest models e.g., Gemma 4